Syllabus
B.Sc. (Information Technology)
(Sem.- VI)

Name of the Course: Data Analytics and Business Intelligence and Computer Security in Practical

Sr.No.	Heading	Particulars
1	**Description the course : Including but Not limited to:**	The subject gives hands-on, laboratory counterpart to the Business Analytics Theory course, offered as a Major Practical in the sixth and final semester of the T.Y.I.T. programme under NEP 2020. Where the theory course builds conceptual and quantitative foundations, this practical course develops the applied, tool-based proficiency that employers directly evaluate in analytics roles. Students graduate having worked on real-world business datasets using three industry-standard platforms: Microsoft Excel, Python/R, and Microsoft Power BI — the exact tool trifecta encountered in Business Analyst, Data Analyst, MIS Executive, and BI Developer job descriptions. The course spans the full analytics pipeline from data warehousing to predictive modelling: importing and analysing data warehouse data in Excel using PivotTables and PivotCharts; performing OLAP-style cube analysis; applying What-If Analysis tools (Goal Seek, Scenario Manager, Data Tables) for business scenario modelling; implementing classification algorithms (Decision Trees, k-Nearest Neighbours) and clustering algorithms (K-Means) using Python (scikit-learn) or R (caret/kmeans); and building Simple and Logistic Regression models on data warehouse datasets. It focuses on lookup-based data retrieval, dimensional modelling (ROLAP, MOLAP, HOLAP cube design with fact and dimension tables), and professional Power BI dashboard development on real-world datasets including Sales data and the Netflix content dataset. Security in Computing Practical builds practical skills in threat detection, encryption, and secure system design through lab simulations. This lab-based course equips students with tools to secure computing environments, covering vulnerabilities, access controls, and ethical hacking basics. It familiarizes students with common cyber threats

		Enable hands-on configuration of firewalls, IDS/IPS, and secure protocols Develop skills in vulnerability scanning and basic penetration testing in controlled environments.
2	**Vertical :**	Major
3	**Type :**	Practical
4	**Credits :**	2 credits (1 credit = 30 Hours of Practical work in a semester)
5	**Hours Allotted :**	60 Hours
6	**Marks Allotted:**	50 Marks
7	**Course Objectives(CO):** CO1. To understand the fundamentals of Data Warehousing including data warehouse architecture, ETL process, OLAP operations, and cube models (ROLAP, MOLAP, HOLAP) CO2. To develop analytical skills using Microsoft Excel by applying Pivot Tables, Pivot Charts, Lookups (VLOOKUP, HLOOKUP), What-If Analysis, and data visualization techniques. CO3. To implement predictive analytics techniques such as Linear Regression and Logistic Regression for real-world datasets. CO4. To design interactive dashboards using Power BI for business decision-making. CO5. To configure and secure network devices using industry-standard protocols. CO6. To design and implement Access Control Lists (ACLs) CO7. To implement Layer 2 security mechanisms on Cisco switches. CO8. To configure IOS Intrusion Prevention Systems (IPS), site-to-site IPsec VPNs and Cisco ASA basic firewall settings using the CLI CO9. To develop host-based security investigation skills using the Windows a. SysInternals Suite and to perform network reconnaissance and attack surface b. mapping using the Nmap port scanner. **CO10.** To capture, filter, and analyse live network traffic using Wireshark, interpreting protocol behaviour across TCP (3-way handshake	

8	**Course Outcomes (OC):** OC 1. Apply data warehousing concepts and perform OLAP analysis using Excel and cube models. OC 2. Perform data analysis and visualization using Pivot tables, What-If analysis, and dashboards. OC 3. Implement data mining techniques such as classification and clustering using R/Python. OC 4. Apply regression techniques (Linear and Logistic) for predictive analysis OC 5. Design dimensional models and create BI dashboards using Power BI. OC 6. Configure routers for Syslog event logging OC 7. Configure, apply, and verify Extended Numbered IPv4 ACLs and IPv6 ACLs OC 8. Configure a Zone-Based Policy Firewall (ZPF) on a Cisco router OC 9. Implement Layer 2 security on Cisco switches OC 10. Configure IOS IPS using the CLI — enabling IPS OC 11. Use the Windows SysInternals Suite to investigate system security OC 12. Capture and analyse live network traffic using Wireshark
9	**Modules:-**
	Module 1:
	1. Perform the analysis for the following: a. Import the data warehouse data in Microsoft Excel and create the Pivot table and Pivot Chart. b. Import the cube in Microsoft Excel and create the Pivot table and Pivot Chart to perform data analysis. 2. Apply the what – if Analysis for data visualization. Design and generate necessary reports based on the data warehouse data. Use Excel. 3. Perform the data classification using classification algorithm using R/Python. 4. Perform the data clustering using clustering algorithm using R/Python. 5. Perform the Linear regression on the given data warehouse data using R/Python. 6. 6. Perform the logistic regression on the given data warehouse data using R/Python. 7. Perform lookups using Microsoft Excel a. Perform vlookup on any products data. b. Perform hlookup on any products data. 8. Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model. 9. Create dashboard in POWER BI of Sales data. Use scattered chart and cards. 10. Create the dashboard in POWER BI of Netflix dataset.
	Module 2:
	21. Configure Routers for a. Syslog, NTP, and b. SSH Operations

	c. OSPF MD5 authentication. 22. Configure AAA Authentication a. Configure a local user account on Router and configure authenticate on the console and vty lines using local AAA b. Verify local AAA authentication from the Router console and the PC-A client 23. Configuring Extended ACLs a. Configure, Apply and Verify an Extended Numbered ACL 24. **Configuring IPv6 ACLs** 25. Configure IP ACLs to Mitigate Attacks a. Verify connectivity among devices before firewall configuration b. Use ACLs to ensure remote access to the routers is available only from management station PC-C. c. Configure ACLs on to mitigate attacks. 26. Configuring a Zone-Based Policy Firewall (ZPF) a. Verify connectivity among devices before firewall configuration. b. Configure a zone-based policy (ZPF) firewall on R3. c. Verify ZPF firewall functionality using ping, SSH, and a web browser 27. DHCP Snooping and ARP Inspection 28. Configure IOS Intrusion Prevention System (IPS)Using the CLI a. Enable IOS IPS. b. Configure logging. c. Modify an IPS signature. d. Verify IPS 29. Implementing Layer 2 security a. Assign the Central switch as the root bridge. b. Secure spanning-tree parameters to prevent STP manipulation attacks. c. Enable port security to prevent CAM table overflow attacks 30. Implementing Layer 2 VLAN Security a. Enable trunking and configure security on the new trunk link 31. Configure and Verify a Site-to-Site IPsec VPN Using CLI 32. Configuring ASA Basic Settings and Firewall Using CLI 33. Exploring Processes, Threads, Handles, and Windows Registry (SysInternals Suite) 34. Use a Port Scanner to Detect Open Ports a. Nmap, a port scanner and network mapping, tool to detect open ports 35. Using Wireshark a. to Observe the TCP 3-Way Handshake b. to Examine a UDP DNS Capture 36. to Examine FTP and TFTP Captures
10	**Text Books**

	5. **Microsoft Excel 365 Bible –** Michael Alexander & Richard Kusleika Publisher: Wiley *(Covers Pivot Tables, What-If Analysis, Lookups, Data Analysis Tools.)* 6. **Python for Data Analysis –** Wes McKinney Publisher: O'Reilly *(Hands-on implementation of regression and data analysis using Python.)* 7. **R for Data Science –** Hadley Wickham & Garrett Grolemund Publisher: O'Reilly(*Practical guide for clustering, classification and visualization using R.)* 8. **Practical Business Intelligence with Power BI –** Brett Powell *Publisher: Apress (Step-by-step dashboard development.)*
11	**Reference Books** 1. **Business Intelligence Guidebook –** Rick Sherman *Practical BI implementation strategies.* 2. **Applied Predictive Modeling –** Max Kuhn & Kjell Johnson *Regression and classification practical modeling.* 3. 3. **Power BI Cookbook –** Brett Powell *Advanced dashboard techniques and case studies.* 4. CCNA Security 2.0 Lab Manual
12	**Internal Continuous Assessment: 40%**
	Continuous Evaluation through: Students are expected to attend each practical and submit the written practical of the previous session. Performing Practical and writeup submission will be continuous internal evaluation. 2.5 marks can be awarded for each practical performance and writeup submission totalling to 50 marks and can be converted to 20 marks.
13	**Semester End Examination: 60%**
	30 marks practical exam of 2 hours duration
14	**Format of Question Paper: (Semester End Examination : 30 Marks. Duration: 2 Hours)** **Practical Slip:** Q1. From Module 1 13 marks Q2. From Module 2 12marks Q3. Journal and Viva 05 marks